Soumiya_Ayyappan_w9_D3

Medium qn

Step 1:







Step 2:

```
01:06 PM (5s)                                                    4

1 from pyspark.sql.functions import when
2 df=df.dropna(subset=["claim_amount"])
3 df=df.withColumn("settlement_gap",col("approved_amount")-col("claim_amount"))
4 df=df.withColumn("review_flag",when(col("claim_amount")>=20000,"HIGH").otherwise("STANDARD"))
5 display(df.count())
6 display(df)
```

See performance (2) ✓ Checked for improvements

df: pyspark.sql.connect.dataframe.DataFrame = [claim_id: string, policy_id: string ... 7 more fields]

9

Table +

	claim_id	policy_id	claim_type	claim_amount	approved_amount	claim_status	claim_date	settlement
1	CL001	P101	Health	18000	15000	APPROVED	2026-09-01	
2	CL002	P102	Vehicle	9000	0	REJECTED	2026-09-01	
3	CL003	P103	Health	25000	20000	APPROVED	2026-09-02	
4	CL005	P105	Vehicle	12000	10000	APPROVED	2026-09-03	
5	CL006	P106	Property	35000	30000	APPROVED	2026-09-03	
6	CL007	P107	Health	8000	0	PENDING	2026-09-04	
7	CL008	P108	Vehicle	22000	18000	APPROVED	2026-09-04	
8	CL009	P109	Property	16000	0	REJECTED	2026-09-05	
9	CL010	P110	Health	28000	24000	APPROVED	2026-09-05	

9 rows | 4.523s runtime Refreshed at 10/14/26, 1:06 PM (2h ago)

Step 3:

```
01:11 PM (4s)                                                    5

1 df.write.format("delta").mode("overwrite").saveAsTable("uc_practice_insurance.claims_ops.claims_silver")
```

See performance (1) ✓ Checked for improvements

```
01:15 PM (3s)                                                    6

1 display(spark.sql("select count(*) as total_count from uc_practice_insurance.claims_ops.claims_silver"))
2 display(spark.sql("select count(*) as claim_count,sum(claim_amount) as total_amount from uc_practice_insurance.claims_ops.claims_silver group by claim_type order
  by claim_type;"))
```

See performance (2) ✓ Checked for improvements

Table +

	total_count
1	9

1 row | 3.296s runtime

Table

	claim_count	total_amount
1	4	79000
2	2	51000
3	3	43000

Step 4:

01:19 PM (3s) 7

```
1 from pyspark.sql.functions import lit
2 df=df.withColumn("source_system",lit("BATCH_CSV"))
3 df.write.format("delta").mode("overwrite").option("mergeSchema","true").saveAsTable("uc_practice_insurance.claims_ops.claims_silver")
```

See performance (1) Checked for improvements

df: pyspark.sql.connect.dataframe.DataFrame = [claim_id: string, policy_id: string ... 8 more fields]

01:19 PM (1s) 8

```
1 df.printSchema()
2 display(df.count())
3 display(df)
```

See performance (2) Checked for improvements

```
root
 |-- claim_id: string (nullable = true)
 |-- policy_id: string (nullable = true)
 |-- claim_type: string (nullable = true)
 |-- claim_amount: integer (nullable = true)
 |-- approved_amount: integer (nullable = true)
 |-- claim_status: string (nullable = true)
 |-- claim_date: date (nullable = true)
 |-- settlement_gap: integer (nullable = true)
 |-- review_flag: string (nullable = false)
 |-- source_system: string (nullable = false)
```

9

Table

	claim_id	policy_id	claim_type	claim_amount	approved_amount	claim_status	claim_date	settlement_
2	CL002	P102	Vehicle	9000	0	REJECTED	2026-09-01	
3	CL003	P103	Health	25000	20000	APPROVED	2026-09-02	
4	CL005	P105	Vehicle	12000	10000	APPROVED	2026-09-03	
5	CL006	P106	Property	35000	30000	APPROVED	2026-09-03	
6	CL007	P107	Health	8000	0	PENDING	2026-09-04	
7	CL008	P108	Vehicle	22000	18000	APPROVED	2026-09-04	
8	CL009	P109	Property	16000	0	REJECTED	2026-09-05	
9	CL010	P110	Health	28000	24000	APPROVED	2026-09-05	

9 rows | 1.158s runtime

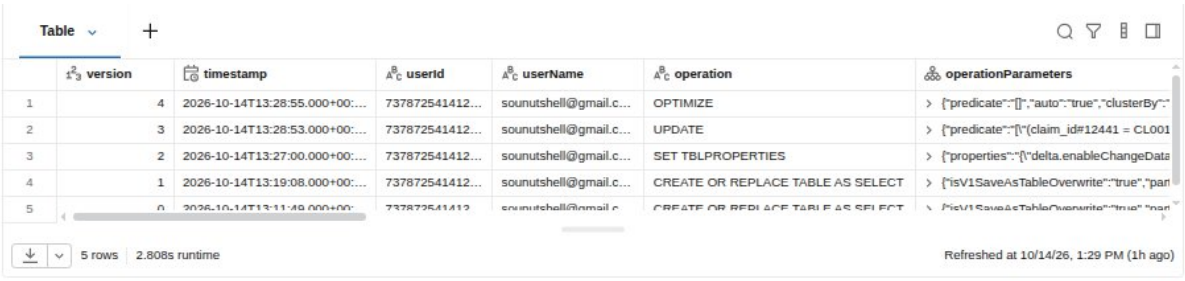
Refreshed at 10/14/26, 1:19 PM (2h ago)

Step 5:


01:26 PM (4s)
9
spark.sql("""
alter table uc_practice_insurance.claims_ops.claims_silver
set tblproperties ('delta.enableChangeDataFeed'= 'true')
""")
See performance (1)
Checked for improvements
DataFrame[]

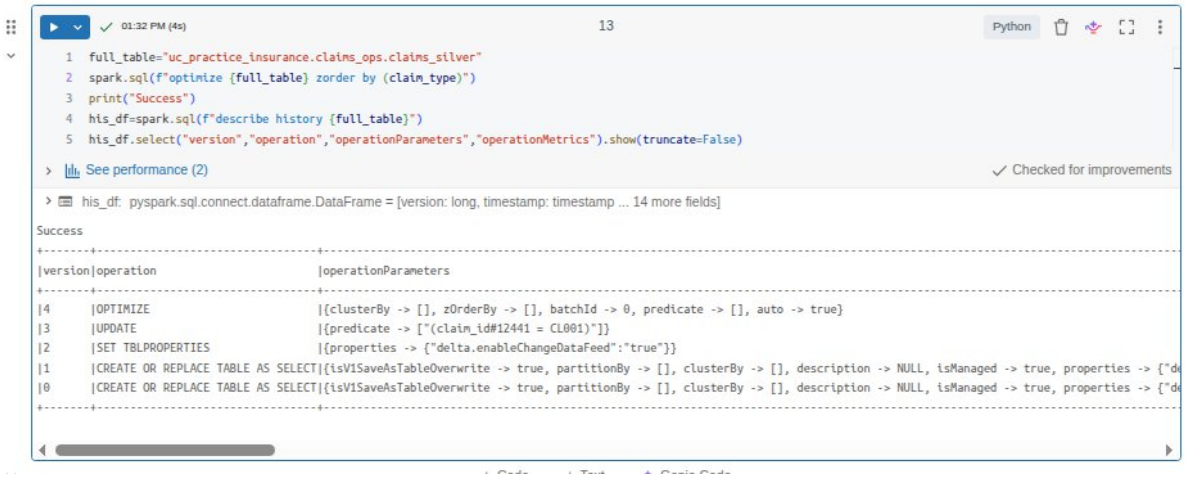

01:28 PM (7s)
10
spark.sql("""
update uc_practice_insurance.claims_ops.claims_silver
set claim_amount=claim_amount+100 where claim_id='CL001'
""")
See performance (1)
Optimize
DataFrame[num_affected_rows: bigint]


01:29 PM (3s)
11
display(spark.sql("describe history uc_practice_insurance.claims_ops.claims_silver"))
See performance (1)
Checked for improvements

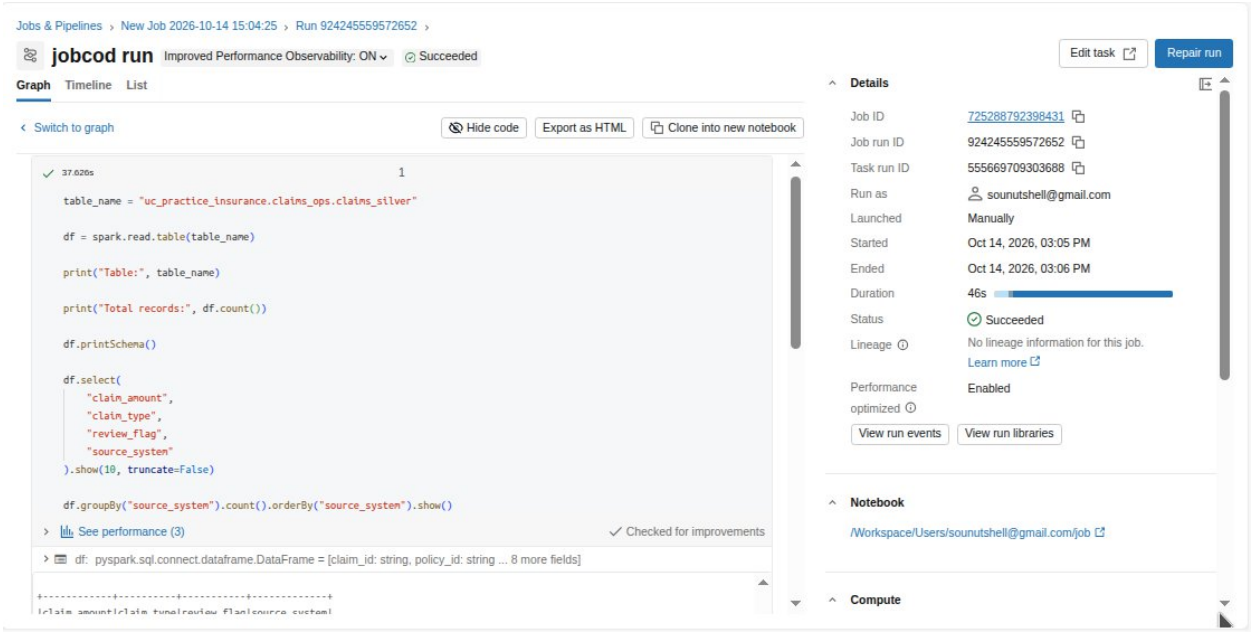

Table
version
timestamp
userId
userName
operation
operationParameters
1 4 2026-10-14T13:28:55.000+00:... 737872541412... sounutshell@gmail.c... OPTIMIZE {"predicate":"[]","auto":"true","clusterBy":"
2 3 2026-10-14T13:28:53.000+00:... 737872541412... sounutshell@gmail.c... UPDATE {"predicate":"[\"(claim_id#12441 = CL001
3 2 2026-10-14T13:27:00.000+00:... 737872541412... sounutshell@gmail.c... SET TBLPROPERTIES {"properties":"{\"delta.enableChangeData
4 1 2026-10-14T13:19:08.000+00:... 737872541412... sounutshell@gmail.c... CREATE OR REPLACE TABLE AS SELECT {"isV1SaveAsTableOverwrite":"true","par
5 0 2026-10-14T13:11:49.000+00:... 737872541412... sounutshell@gmail.c... CREATE OR REPLACE TABLE AS SELECT {"isV1SaveAsTableOverwrite":"true","par
5 rows | 2.808s runtime
Refreshed at 10/14/26, 1:29 PM (1h ago)


01:31 PM (2s)
12
%sql
select * from uc_practice_insurance.claims_ops.claims_silver version as of 2 where claim_id='CL001';
See performance (1)
Checked for improvements
Table
claim_id
policy_id
claim_type
claim_amount
approved_amount
claim_status
claim_date
settlement_g
1 CL001 P101 Health 18000 15000 APPROVED 2026-09-01

Step 6:



Jobs and pipelines



```
+------------+----------+-----------+-------------+
|claim_amount|claim_type|review_flag|source_system|
+------------+----------+-----------+-------------+
|18100       |Health    |STANDARD   |BATCH_CSV    |
|9000        |Vehicle   |STANDARD   |BATCH_CSV    |
|25000       |Health    |HIGH       |BATCH_CSV    |
|12000       |Vehicle   |STANDARD   |BATCH_CSV    |
|35000       |Property  |HIGH       |BATCH_CSV    |
|8000        |Health    |STANDARD   |BATCH_CSV    |
|22000       |Vehicle   |HIGH       |BATCH_CSV    |
|16000       |Property  |STANDARD   |BATCH_CSV    |
|28000       |Health    |HIGH       |BATCH_CSV    |
+------------+----------+-----------+-------------+

+-------------+-----+
|source_system|count|
+-------------+-----+
|    BATCH_CSV|    9|
+-------------+-----+
```